⚡ QuickLifeNow Enterprise SaaS Platform • Version 2.0 Architectural Specification

Complete System Architecture & Feature Workflow Flowcharts

An authoritative, enterprise-grade flowchart specification demonstrating the full lifecycle, branching logic, role-based governance, and real-time state machine of the QuickLifeNow Task Management platform.

📌 Executive Answer: How Many Flowcharts Are Necessary to Completely Understand the Project?

To understand 100% of the QuickLifeNow platform without missing any technical or business nuance, exactly **7 focused flowcharts** are required: **1 Master Executive Blueprint** representing the end-to-end bird's-eye view, plus **6 Deep-Dive Domain Flowcharts** detailing every operational subsystem (Creation/Routing, Clarification Loop, Quality Review & Revision Loop, Multi-branch Hierarchical Delegation, Governance & Oversight, and RBAC Security Isolation).

DIAGRAM NO.	SYSTEM WORKFLOW DOMAIN	KEY BUSINESS & LOGIC COVERED	CORE PLATFORM STAKEHOLDERS
Diagram 1	**Master End-to-End System Lifecycle**	Complete lifecycle from creation to sign-off and analytics archive	C-Level, PMO, All Roles
Diagram 2	**Task Creation, Routing & Allocation**	Handling methods (Work Alone, Assign, Collaborate), auto-start date	Creators, Dept Heads
Diagram 3	**Execution & Missing Details Loop**	Assignee questions, state transition to post_comment, assigner response	Assignees, Assigners
Diagram 4	**Review, Quality Gate & Revision Loop**	Deliverable submission, review queue, rejection/return & resubmission	Reviewers, Assignees
Diagram 5	**Hierarchical Subtask Branching Tree**	Cascading delegation (can_assign_further), milestone rollups	Operations Managers, Team Leads

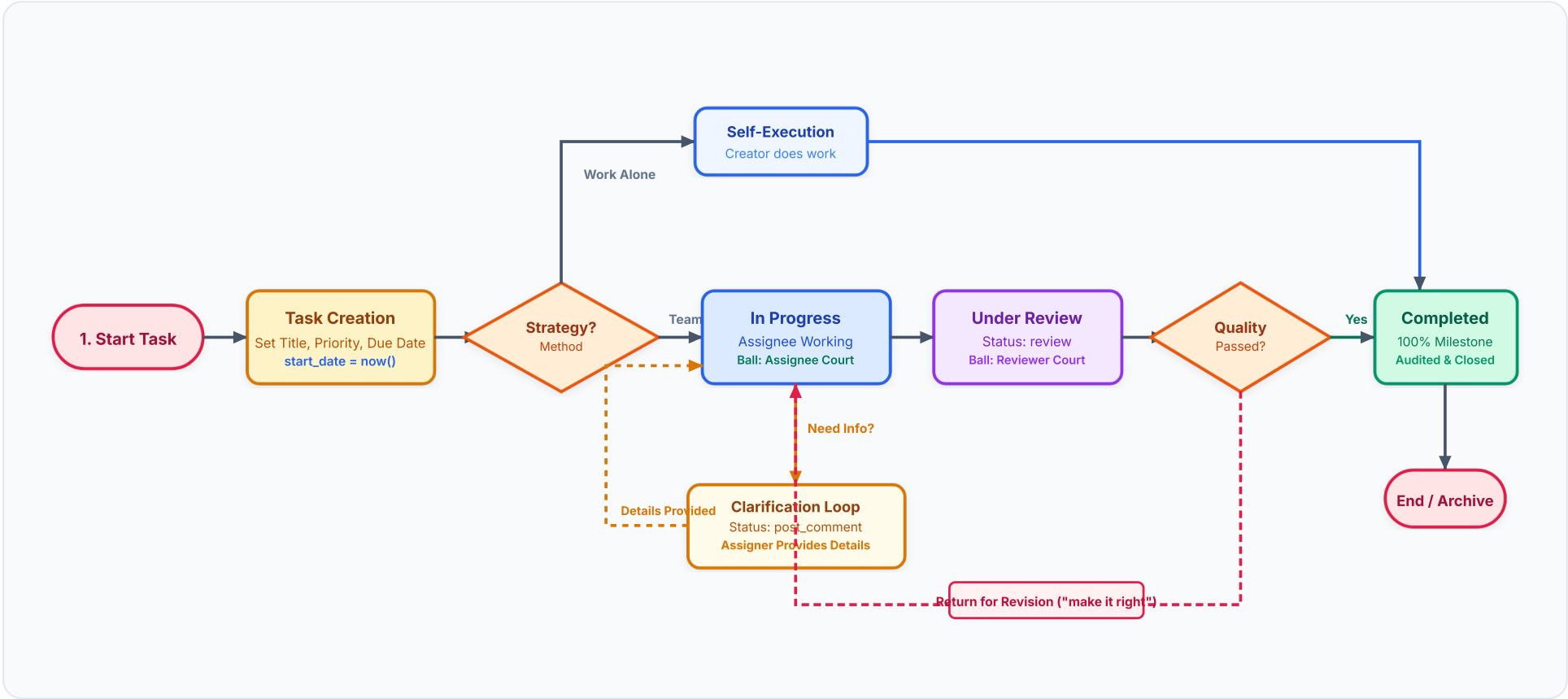
DIAGRAM NO.	SYSTEM WORKFLOW DOMAIN	KEY BUSINESS & LOGIC COVERED	CORE PLATFORM STAKEHOLDERS
Diagram 6	**Governance, Hold & Cancellation**	Put on hold, resume, soft cancellation, deletion authorization	Administrators, Auditors
Diagram 7	**RBAC Security & Privacy Isolation**	3-Tier RBAC, peer privacy protection, audit logging, API security	Security Team, System Architects

ARCHITECTURAL BLUEPRINT

1. Master End-to-End Task Lifecycle Flowchart

A comprehensive bird's-eye view showing the complete lifecycle of tasks from initiation through execution, quality gates, sign-off, and archival.

Standard Notation: Terminator (Start / End) · Action / Process · Decision Gate · Data / I/O Document · Auto-reversion & Return Loops



- **REAL-TIME BALL TRACKER**
- **STRICT REJECTION ISOLATION**
- **MILESTONE CONVERGENCE**

The system dynamically moves the **"Ball is in YOUR court"** indicator: In Progress → Assignee, Review → Reviewer, Missing Details → Assigner, Revision → Assignee.

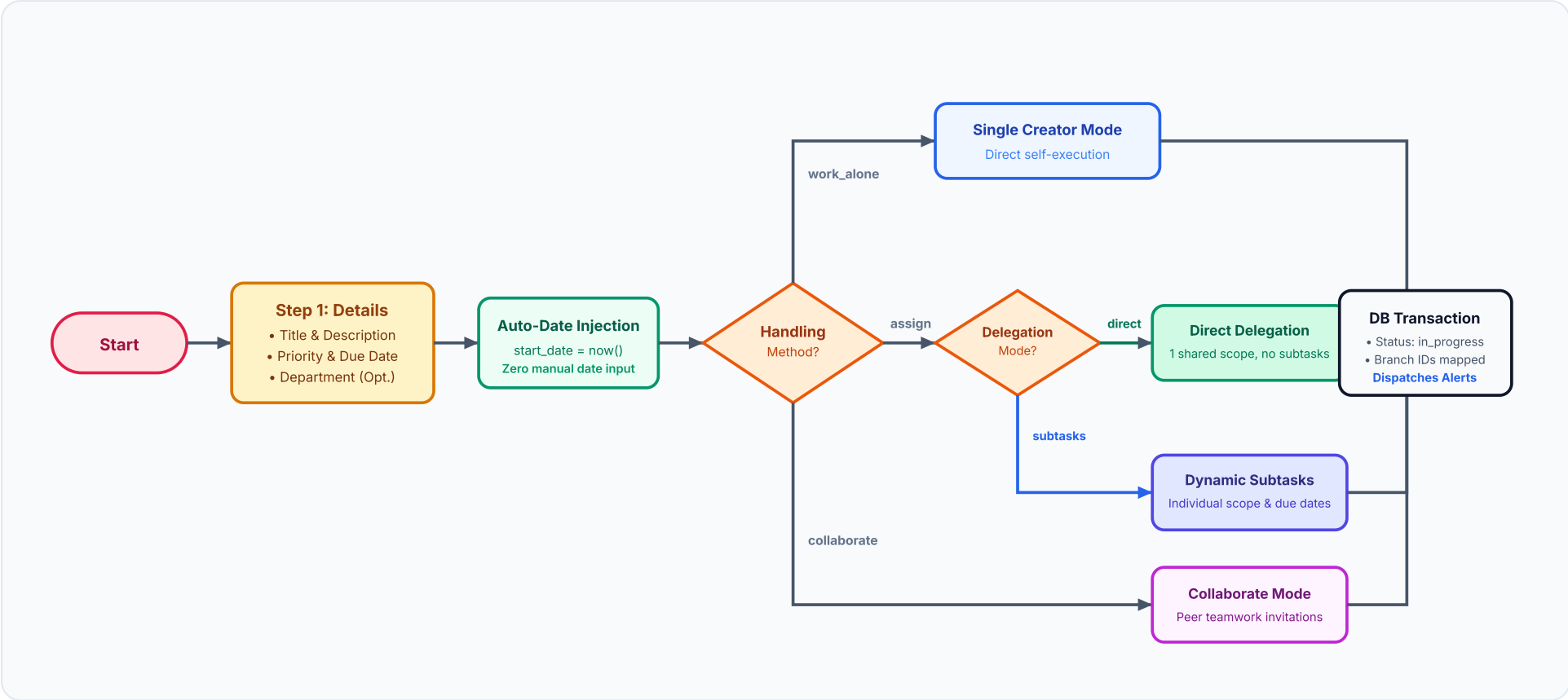
When a reviewer returns work, past clarification requests are superseded immediately. The task transitions to `post_comment` awaiting assignee fixes.

Child subtasks run concurrently with peer privacy. The parent task automatically unlocks Final Creator Sign-off only when 100% of branches are approved.

MODULE 1 ARCHITECTURE

2. Task Creation, Routing & Allocation Flowchart

Demonstrates how tasks are configured, auto-dated on creation, routed through departments, and split into direct or multi-branch subtasks.



- **AUTOMATIC START TIMESTAMP**

Manual start date inputs have been eliminated. Tasks and subtasks automatically store `start_date = now()` on creation, preventing discrepancies.

- **DUAL DELEGATION MODES**

Creators choose between **Direct Delegation** (one shared project deliverable) or **Subtask Scopes** (dedicated cards and private scopes per employee).

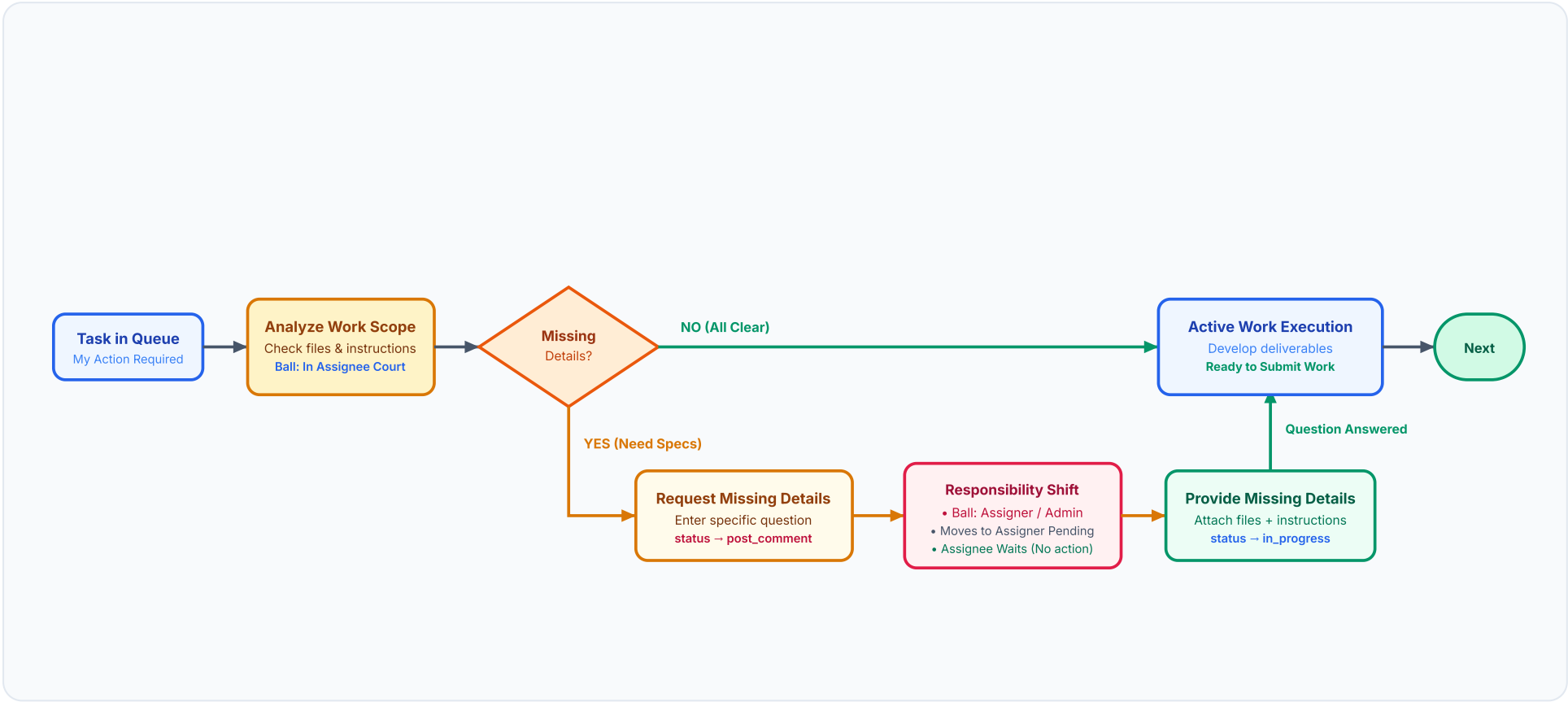
- **SELF-ASSIGNMENT PREVENTION**

FormRequest validation strictly blocks assigning a task to yourself under `assign` mode, directing users to `work_alone` for self-contained work.

MODULE 2 ARCHITECTURE

3. Task Execution & Missing Details Loop Flowchart

Illustrates how team members request clarification, shift real-time responsibility, and how assigners provide missing assets without breaking project flow.



- **CONTEXTUAL BANNERS**

When an employee asks a question, the top bar renders an amber alert `"Missing Details Requested"` with an instant `[Provide Missing Details]` modal for the assigner.

- **QUEUE HANDOVER AUTOMATION**

The task moves out of the employee's active queue and into the assigner's **"My Action Required"**, ensuring managers never lose track of questions blocking employees.

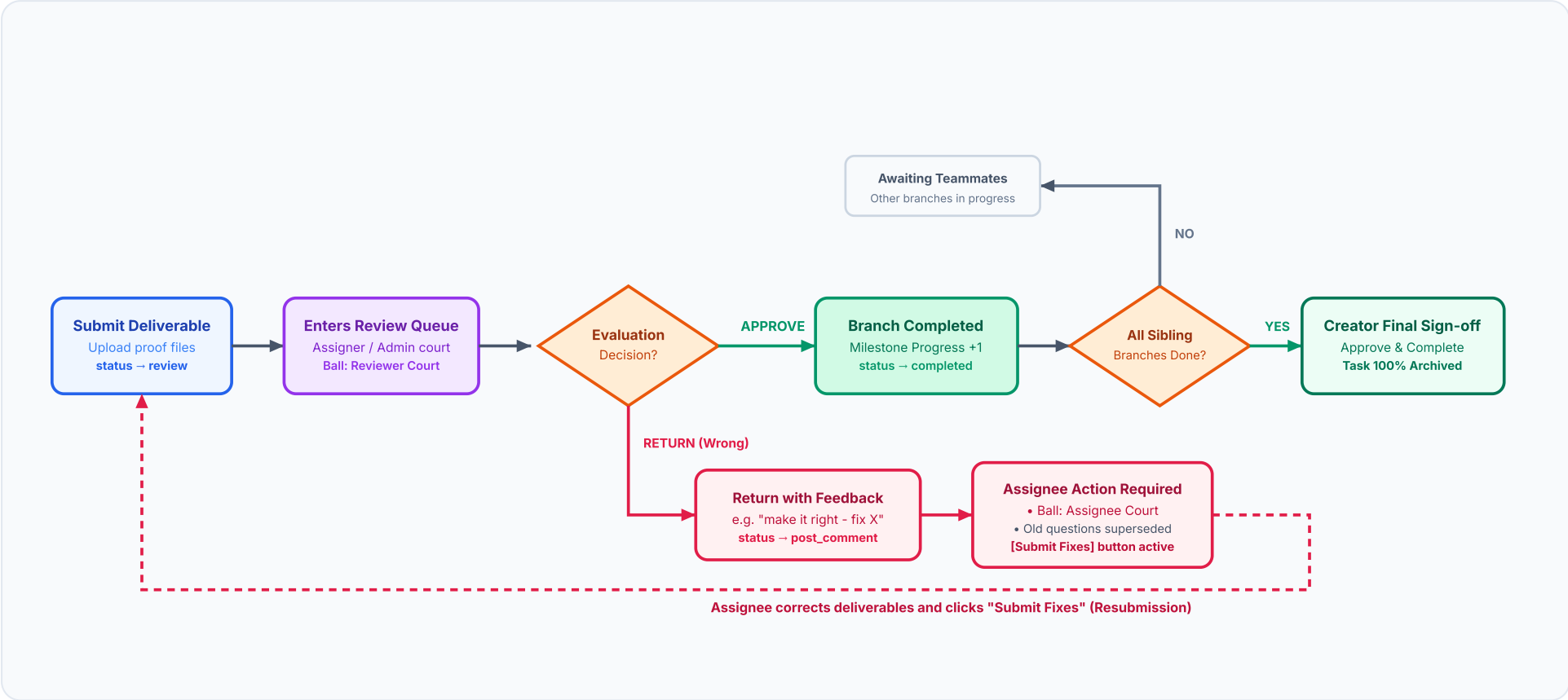
- **AUTOMATIC RESOLUTION STATE**

Submitting clarification details automatically flips the task back to `in_progress` and marks the question answered, returning the ball to the assignee's court immediately.

MODULE 3 ARCHITECTURE

4. Deliverable Submission, Review & Revision Return Loop

Details how submitted deliverables are evaluated, how returns for correction cleanly route feedback, and how resubmissions pass through quality gates.



- **CLEAN FEEDBACK ROUTING**

When Admin rejects work with remarks (e.g. *"make it right"*), old questions are superseded. The employee receives a red banner with notes and a `[Submit Fixes]` action.

- **CONFLICT-FREE DASHBOARD**

Admin dashboard displays *"Returned for revision • Awaiting assignee"* with zero false alerts, while the employee's screen clearly highlights *"Action Required"*.

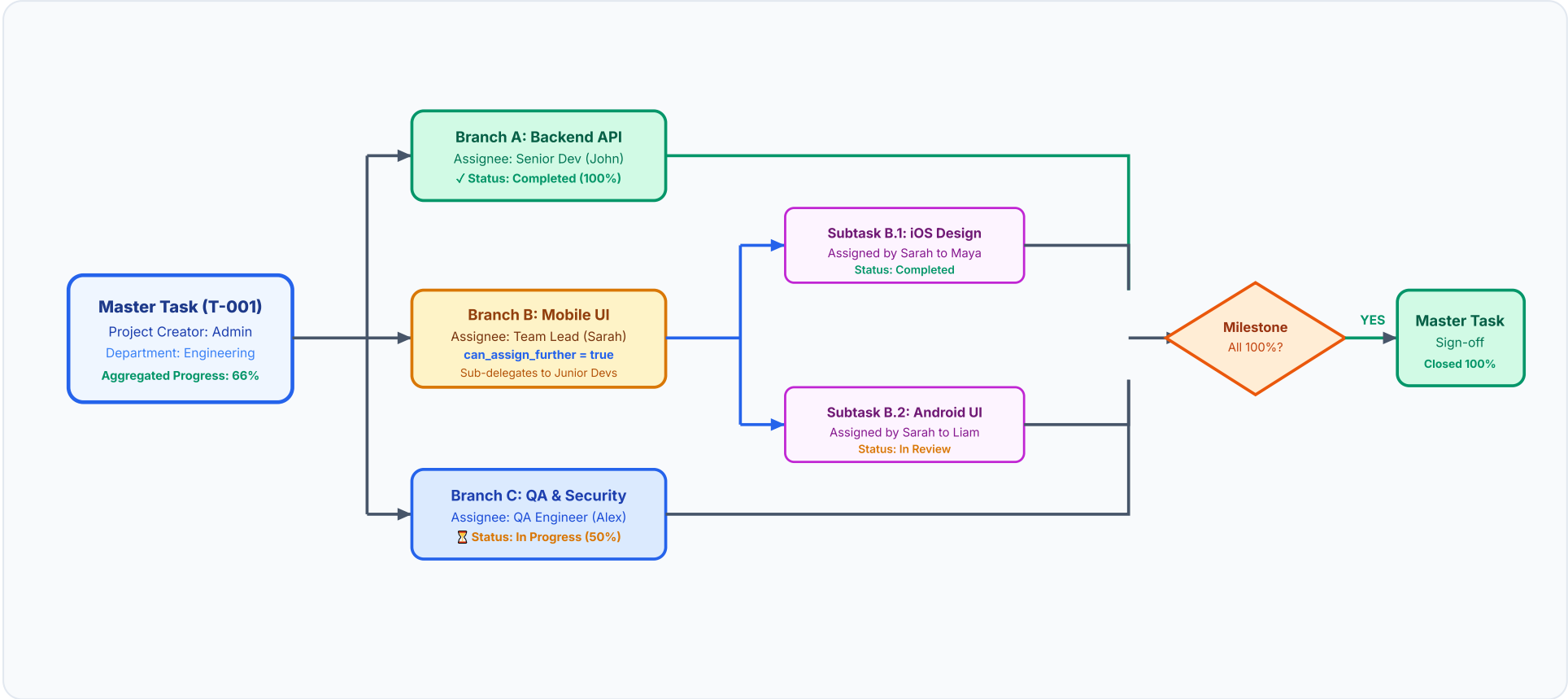
- **DUAL-TIER SIGN-OFF GATE**

Individual subtasks are approved first by their respective assigners. Once all sibling branches succeed, the Project Creator performs the final master sign-off.

MODULE 4 ARCHITECTURE

5. Hierarchical Subtask Branching & Milestone Convergence

Illustrates multi-level organizational delegation down manager trees, peer privacy boundaries, and dynamic milestone progress aggregation.



- **MULTI-TIER SUB-DELEGATION**

When enabled (`can_assign_further = true`), assignees can branch scopes down their organization tree. Sub-assignees report directly up their delegation hierarchy.

- **STRICT PEER PRIVACY ISOLATION**

Assignees on Branch A cannot inspect private deliverables, files, or review comments on Branch B or C, ensuring confidential deliverables remain segregated.

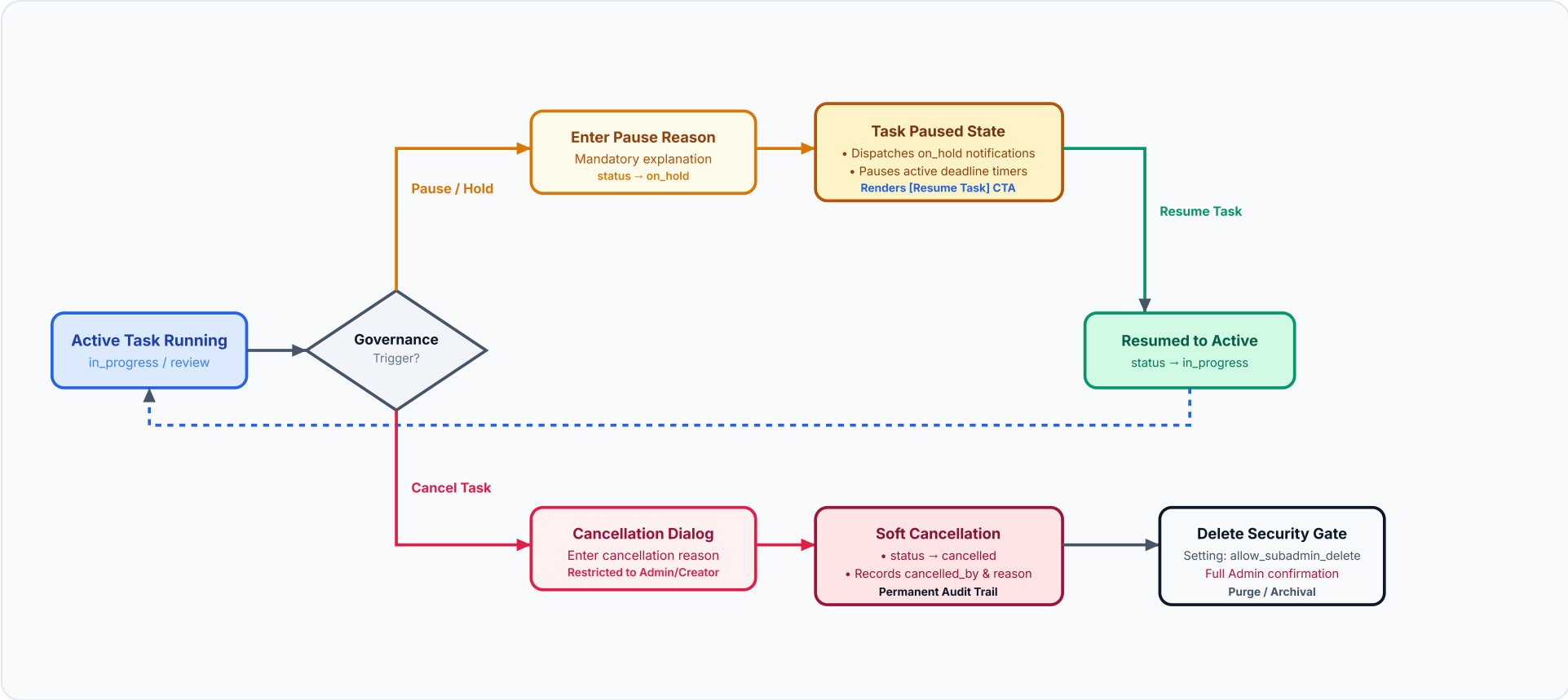
- **AUTOMATIC MILESTONE AGGREGATION**

The master progress bar mathematically aggregates child branches: `Progress = (Completed Branches / Total Branches) * 100`, ensuring real-time executive reporting.

MODULE 5 ARCHITECTURE

6. Administrative Governance, On-Hold & Cancellation Flowchart

Covers administrative oversight, pausing project SLA clocks with reasons, soft cancellation with audit trails, and deletion security gates.



- **SLA PAUSE & RESUME**

Putting a task on hold requires documenting a clear rationale. All SLA timers pause and notification banners keep assignees informed until resumed.

- **TRACEABLE SOFT CANCEL**

Cancelled tasks preserve all historical files, comments, and milestones with the exact timestamp and authorizing user recorded in the database.

- **DELETION AUTHORIZATION**

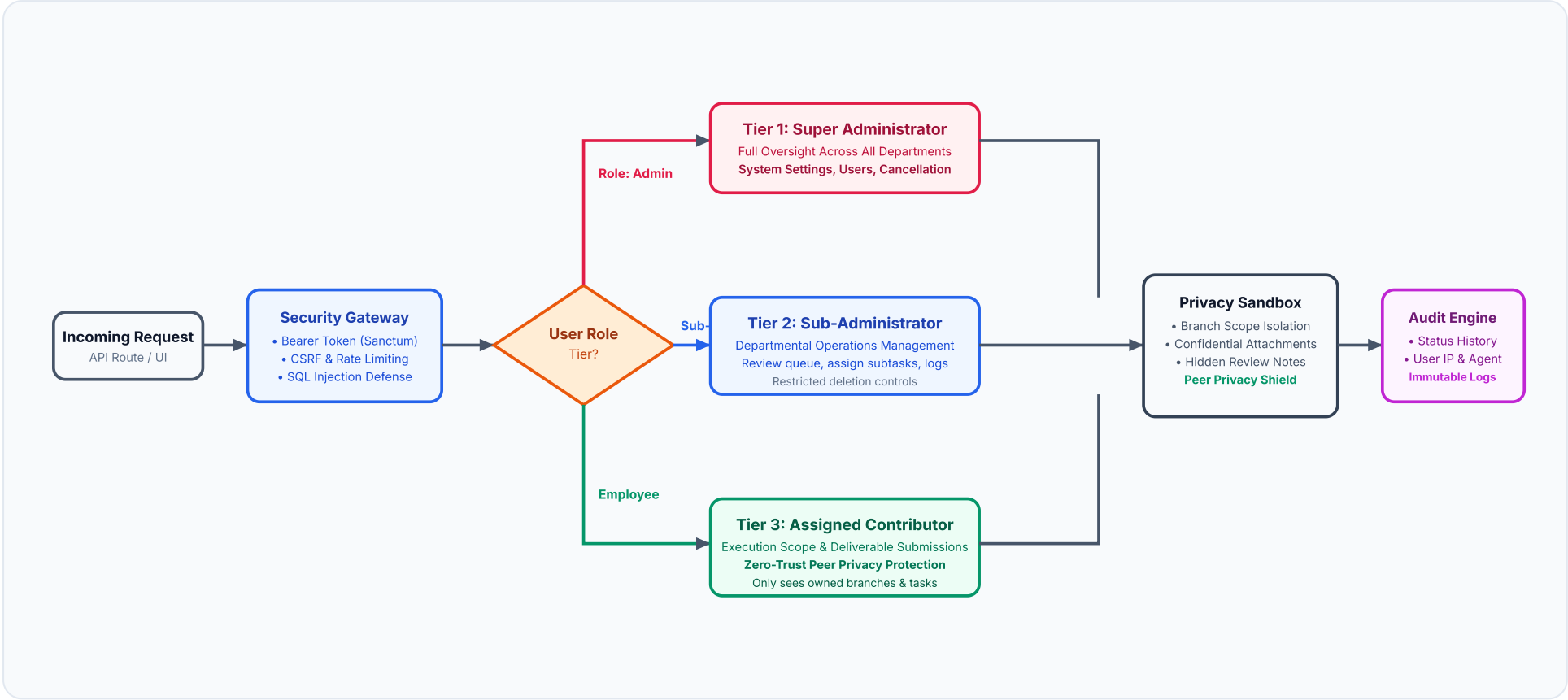
Destructive deletions require Full Administrator clearance or an explicit system toggle

(`allow_subadmin_delete_records`), preventing accidental data loss.

SECURITY ARCHITECTURE

7. Role-Based Access Control (RBAC) & Security Architecture

Details the 3-tier user authority matrix, API request security boundaries, peer privacy isolation, and compliance audit trail pipelines.



- **3-TIER ENTERPRISE RBAC**

Enforces clear separation of concerns between **Admin** (Executive Governance), **Sub-Admin** (Operational Oversight), and **Employee** (Focused Execution).

- **ZERO-TRUST PEER PRIVACY**

Employees assigned to collaborative tasks cannot inspect each other's private subtask descriptions, uploaded files, or reviewer feedback unless explicitly shared.

- **TAMPER-PROOF AUDIT TRAIL**

Every milestone transition, comment, review decision, and file attachment is permanently recorded in

`task_status_history` with user attribution for regulatory compliance.